

A Deep Learning Model for the Accurate and Reliable Classification of Disc Degeneration Based on MRI Data

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Publication / Venue:	Investigative Radiology
Year:	2021
DOI:	10.1097/rli.0000000000000709
Primary Source URL:	https://pubmed.ncbi.nlm.nih.gov/32694451/
Inventory Category:	Disc degeneration / herniation
Comparability / Priority:	High Priority: Critical

Study Details & Methodology from Literature Inventory

Study Type / MRI View:	Deep-learning grading Lumbar MRI
Dataset & Cohort:	None (Sample: 1,599 patients / 7,948 discs reported in later review)
Disease / Target:	Intervertebral disc degeneration
Model / AI Method:	Deep CNN
Evaluation Metrics:	None
Key Finding / Relevance:	None
Thesis Context (Ch 2):	Large disc-level degeneration-grading study; useful contrast with stenosis-based RSNA labels.

Abstract / Summary

Objectives Although magnetic resonance imaging–based formalized grading schemes for intervertebral disc degeneration offer improved reproducibility compared with purely subjective ratings, their intrarater and interrater reliability are not nearly good enough to be able to detect small to medium effects in clinical longitudinal studies. The aim of this study thus was to develop a method that enables automatic and therefore reproducible and reliable evaluation of disc degeneration based on conventional clinical image data and Pfirrmann's grading scheme. **Materials and Methods** We propose a classifier based on a deep convolutional neural network that we trained on a large, manually evaluated data set of 1599 patients (7948 intervertebral discs). To improve upon the status quo, we focused on the quality of the training data and performed extensive hyperparameter optimization. We assessed the potential benefits of optimizing loss functions beyond common cross-entropy loss, such as soft kappa loss, ordinal cross-entropy loss, or regression losses. We furthermore experimented with ways to mitigate class imbalance by pooling classes or using class-weighted loss functions. During model development and hyperparameter optimization, we used a fixed 90%/10% training/validation set split. To estimate real-world prediction performance, we performed 10-fold cross-validation. **Results** The evaluated image data results in a Gaussian degeneration grade distribution, and thus grades 1 and 5 are slightly underrepresented in the training set. Our default cross-entropy–based classifier achieves a reliability of $\kappa = 0.92$ (Cohen κ), an average sensitivity of 90.2%, and an average precision of 92.5%. In 99.2% of validation cases, the network's prediction deviates at most 1 Pfirrmann grades from the ground truth. Framed as an ordinal regression problem, the mean absolute error between the ground truth and the prediction is 0.08 Pfirrmann grade with a correlation of $r = 0.96$. The results of the 10-fold cross validation confirm those performance estimates, indicating no substantial overfitting. More sophisticated loss functions, class-based loss weighting, or class pooling did not lead to improved classification performance overall. **Conclusions** With a reliability of $\kappa > 0.9$, our system clearly outperforms average human interrater as well as intrarater reliability. With an average sensitivity of more than 90%, our classifier also surpasses state-of-the-art machine learning solutions for automatically grading disc degeneration.